
PROJECT DOCUMENTATION

[DRIVING LICENCE PORTAL]

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Table of Contents

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TABLE OF CONTENTS

Cover Page	i
Certificate	ii
Acknowledgement	iii
Abstract	iv
Table of Contents	v
List of Figures	vi
List of Tables	vii
Chapter 1: Introduction	1
1.1 Problem Statement	2
1.2 Objectives	3
1.3 Scope	4
Chapter 2: Requirements Analysis	5
2.1 Functional & Non-Functional Requirements	6
2.2 Use Case Summary	8
Chapter 3: System Design	10
3.1 System Architecture	11
3.2 Component Hierarchy	13
3.3 Route Map	15
3.4 UI Wireframes	17
Chapter 4: Implementation	22
4.1 Technology Stack	23
4.2 Key Modules	25
4.3 Challenges and Solutions	28
Chapter 5: Testing	30
5.1 Test Cases	31
5.2 Test Results	33
Chapter 6: Results and Discussion	35
6.1 Results	36

6.2 Discussion	38
Chapter 7: Conclusion and Future Work	40
7.1 Conclusion	41
7.2 Lessons Learned	42
7.3 Future Enhancements	43
References	44
Appendix	45

Abstract

This project focuses on digitizing key transport services under the MeeSeva government services platform, including learner's licence applications, driving test scheduling, driving licence issuance, and renewals. Traditional processes often require citizens to visit RTO offices multiple times, resulting in long queues, delays, and limited transparency.

The proposed solution is a web-based application developed using React.js, Node.js, and Express.js that streamlines the complete driving licence workflow through a single digital platform. Users can verify eligibility, submit applications, schedule driving tests, book available slots, make mock fee payments, track application status in real time, and download certificates in PDF format.

The frontend provides a responsive and user-friendly interface, while REST APIs handle application processing and communication. Optional MySQL integration supports persistent storage and administration functions. The project demonstrates how modern frontend engineering frameworks and UI technologies can improve efficiency, reduce manual errors, and enhance citizen convenience in government service delivery.

Keywords: React.js, Node.js, Express.js, REST APIs, SPA, Routing, State Management, MeeSeva Portal

Chapter 1: Introduction

1.1 Problem Statement

Citizens applying for driving licences often face lengthy manual procedures, multiple visits to RTO offices, and difficulty tracking application progress. Existing processes lack transparency and require significant paperwork. There is a need for a centralized digital platform that simplifies learner's licence applications, test scheduling, fee payments, and licence issuance. A React.js-based Single Page Application provides a responsive and efficient solution for delivering these services online.

1.2 Objectives

The objectives of this project are:

- To automate driving licence eligibility verification and application processing.
- To provide real-time status tracking, notifications, and certificate generation

1.3 Scope

The project covers learner's licence applications, eligibility checking, driving test scheduling, slot booking, mock fee payment, status tracking, certificate download, and administration functions. The system provides a responsive web interface using React.js and communicates with backend services through REST APIs. Real government payment gateways and official transport databases are not integrated and are simulated for educational purposes.

Chapter 2: Requirements

2.1 Functional & Non-Functional Requirements

The table below lists the key requirements of the system:

ID	Type	Requirement	Priority	Category
R-01	Functional	Users can navigate between pages without a full page reload (SPA routing).	High	Routing
R-02	Functional	Dynamic content displayed based on URL parameters (e.g., /product/:id).	High	Routing
R-03	Functional	Certain pages restricted to authenticated users (protected routes).	High	Routing
R-04	Functional	All form fields validated with real-time feedback and clear error messages.	High	Forms
R-05	Functional	Application is keyboard-navigable and screen-reader accessible (WCAG 2.1 AA).	Medium	Accessibility
R-06	Functional	Heavy components loaded on demand to reduce initial bundle size (lazy loading).	Medium	Performance
R-07	Functional	Users can apply for licences, book test slots, track status, and download certificates through a unified portal.	High	Application Management
R-08	Non-Functional	Initial page load must complete within 3 seconds on a standard 4G connection.	High	Performance
R-09	Non-Functional	UI must be responsive across desktop (1280px+), tablet (768px), and mobile (375px).	High	Usability
R-10	Non-Functional	All text must meet WCAG 2.1 AA contrast ratio (minimum 4.5:1 for normal text).	Medium	Accessibility

2.2 Use Case Summary

Use Case 1: Apply for Learner's Licence

Actor: Citizen

Precondition: User is registered and logged in.

Main Flow:

Fill application form.

Submit required details.

Eligibility is verified.

Application is saved.

Outcome: Learner's licence application is successfully submitted.

Use Case 2: Schedule Driving Test

Actor: Citizen

Precondition: Eligible application exists.

Main Flow:

View available slots.

Select preferred date and time.

Confirm booking.

Outcome: Driving test slot is reserved.

Use Case 3: Track Application Status

Actor: Citizen

Precondition: Application submitted.

Main Flow:

Open dashboard.

View application status.

Receive notifications.

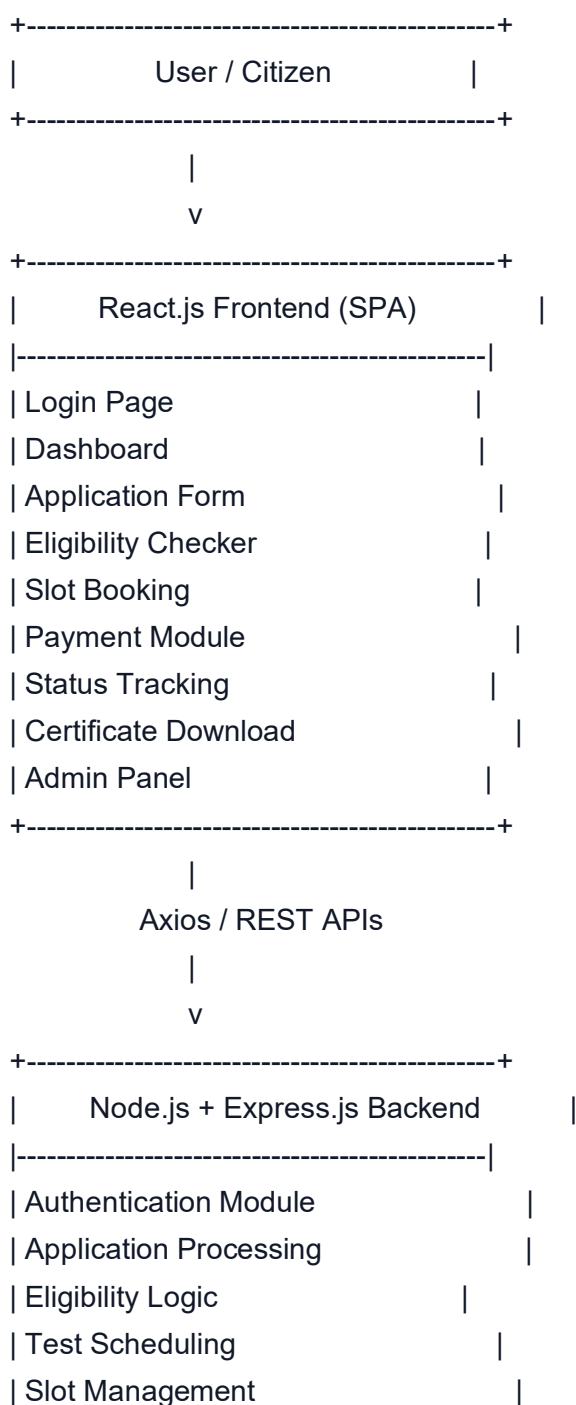
Outcome: User stays informed throughout the process

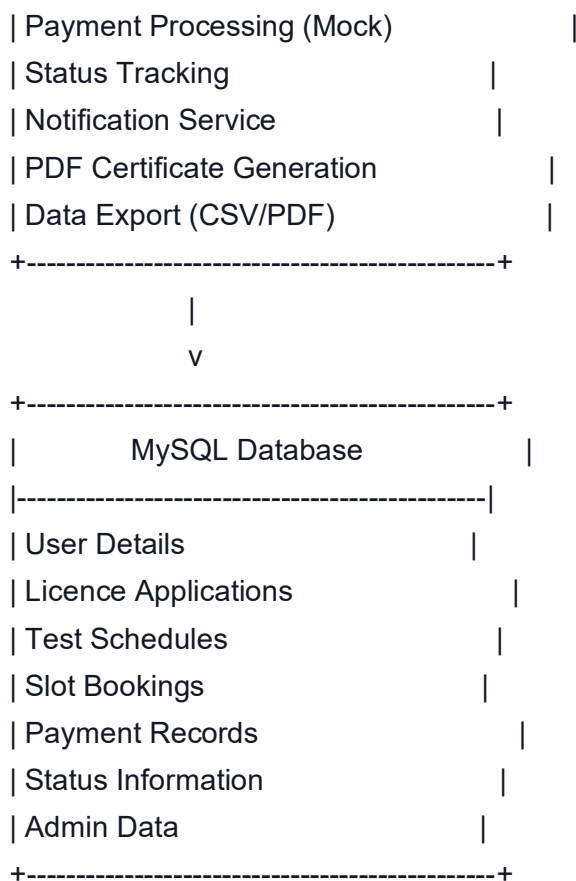
Chapter 3: System Design

3.1 System Architecture

The MeeSeva Driving Licence Portal follows a **three-tier architecture** consisting of the Presentation Layer (Frontend), Application Layer (Backend), and Data Layer (Database). The frontend is developed using React.js and provides an interactive Single Page Application (SPA) for users. The backend is built using Node.js and Express.js, which process requests, implement business logic, and expose REST APIs. Data is stored in a MySQL database, while Axios is used for communication between the frontend and backend. The architecture supports user authentication, application processing, eligibility verification, slot booking, payment handling, status tracking, certificate generation, and administrative operations.

Figure 3.1: System Architecture Diagram





3.2 Component Hierarchy

Component	Responsibility
App.jsx	Root component —managing routes
LoginPage.jsx	User authentication
Dashboard.jsx	Displays application and licence status
ApplicationForm.jsx	Licence application submission
EligibilityChecker.jsx	Eligibility verification
TestScheduler.jsx	Driving test scheduling
SlotBooking.jsx	Slot selection and confirmation
PaymentPage.jsx	Mock fee payment
StatusTracker.jsx	Application tracking
CertificateDownload.jsx	PDF certificate generation
AdminPanel.jsx	Application management and reporting

3.3 Route Map

Path	Component	Access
/	HomePage	Public
/login	LoginPage	Public
/dashboard	Dashboard	Protected (auth required)
/apply	ApplicationForm	Protected
/eligibility	EligibilityChecker	Protected
/schedule	TestScheduler	Protected
/payment	PaymentPage	Protected
/status	StatusTracker	Protected
/certificate	CertificateDownload	Protected
/admin	AdminPanel	Protected

3.4 UI Wireframes

[Insert wireframe images of your key screens. Add a figure caption for each. Minimum: Home page, a form page, and the dashboard.]

Wireframe 1 – Login Page

MeeSeva Portal

MeeSeva Government Logo

Mobile Number

Password

Sign In

Aadhaar Login

Aadhaar Number

Send OTP

New User? Register Here

Figure 3.2: Wireframe – Login Page



Wireframe 2 – Dashboard

Header: MeeSeva Driving Licence Portal

Dashboard

Active Applications : 2

Licence Status : Active

Next Test Date : 14-Jun

Fees Paid : ₹800

Quick Actions

Apply

Track

Download

Figure 3.3: Wireframe – Dashboard Page

Wireframe 3 – Driving Licence Application Form

Driving Licence Application Form

Personal Information

Name

DOB

Gender

Mobile Number

Aadhaar Number

Address Details

Address

City

State

Pincode

Previous

Next

Figure 3.4: Wireframe – Licence Application Form

Wireframe 4 – Eligibility Check Page

Eligibility Checker

Date of Birth

Licence Type

LMV ▼

Check Eligibility

Result

✓ Eligible for Learner's Licence

Minimum Age Requirement Satisfied

Figure 3.5: Wireframe – Eligibility Verification

Wireframe 5 – Slot Booking Page

Slot Booking

Select Test Center

Hyderabad RTO ▼

Select Date

20-06-2026 ▼

Available Slots

10:00 AM

11:00 AM

02:00 PM

03:00 PM

Confirm Booking

Figure 3.6: Wireframe – Driving Test Slot Booking

Wireframe 6 – Status Tracking Page

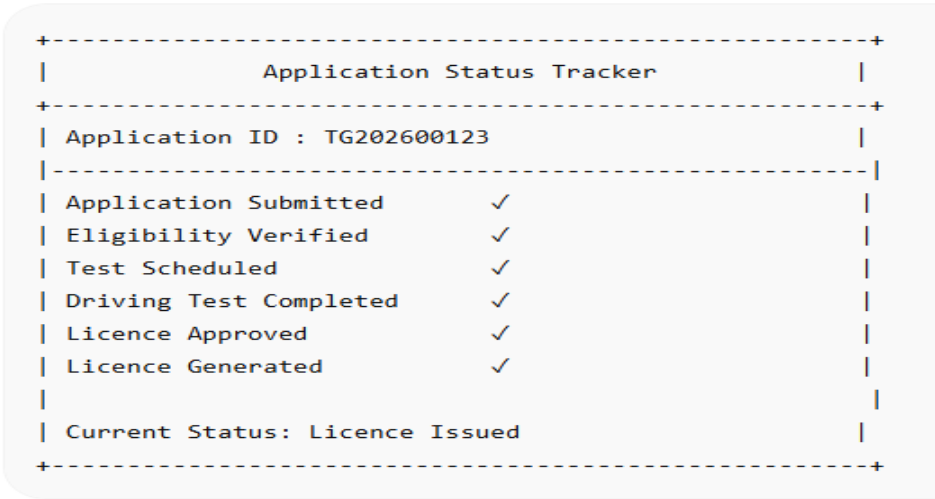


Figure 3.7: Wireframe – Application Status Tracking

Wireframe 7 – Certificate Download Page



Figure 3.8: Wireframe – Certificate Download Screen

Figure 3.2: Wireframe — Home Page

Chapter 4: Implementation

4.1 Technology Stack

Category	Technology	Purpose
UI Library	React.js 18	Component-based SPA development
Routing	React Router v6	Client-side navigation and protected routes
Build Tool	Vite	Fast development server and bundling
Styling	Bootstrap / Tailwind CSS	Responsive UI
State Management	Context API / Redux	State Handling
API / Data	Axios, REST APIs	Communication
Version Control	Git & GitHub	Source control

4.2 Key Modules

4.2.1 Routing Module

The User Authentication module allows citizens to securely log into the MeeSeva Driving Licence Portal using their registered mobile number or Aadhaar-based authentication. The module validates user credentials and grants access to protected pages such as the dashboard, application form, and status tracking. React Router is used to restrict unauthorized access while Axios communicates with the backend authentication APIs.

Key Features

- Mobile Number Login
- Aadhaar Login (OTP Simulation)
- User Validation
- Protected Routes
- Session Management

4.2.2 Form Engineering

The Licence Application Module enables citizens to apply for a learner's licence or permanent driving licence through an online form. Users enter their personal details, address, identity information, and licence category. React controlled forms validate all mandatory fields before submitting the application to the backend

[Paste a key code snippet from your form component or validation hook.]

```
function submitApplication() {  
  
    if(name === "" || dob === "" || aadhaar === ""){  
        alert("Please fill all required fields");  
        return;  
    }  
  
    alert("Application Submitted Successfully");
```

```
}
```

4.2.3 Accessibility

The following accessibility measures were implemented:

The Eligibility Verification Module determines whether the applicant satisfies the age requirement for obtaining a learner's or permanent driving licence. Based on the applicant's Date of Birth and selected licence type, the system instantly displays the eligibility result.

Working:

1. User selects Date of Birth.
2. User selects Licence Type.
3. System calculates the applicant's age.
4. Eligibility is verified.
5. Result is displayed immediately.

4.2.4 Performance Optimizations

The Slot Booking Module enables applicants to select their preferred RTO, examination date, and available time slot for the driving test. The system displays only available slots and confirms successful booking after user confirmation.

Working:

1. User selects RTO office.
2. User selects available test date.
3. User selects preferred time slot.
4. Booking is confirmed.
5. Confirmation message is displayed.

4.3 Challenges and Solutions

Challenge	Solution Adopted
Managing multiple application modules	Modular React component architecture
Real-time status updates	REST API integration and state management
Slot booking conflicts	Validation and scheduling logic
Data export and reporting	CSV/PDF generation modules

Chapter [Module]

5: Testing

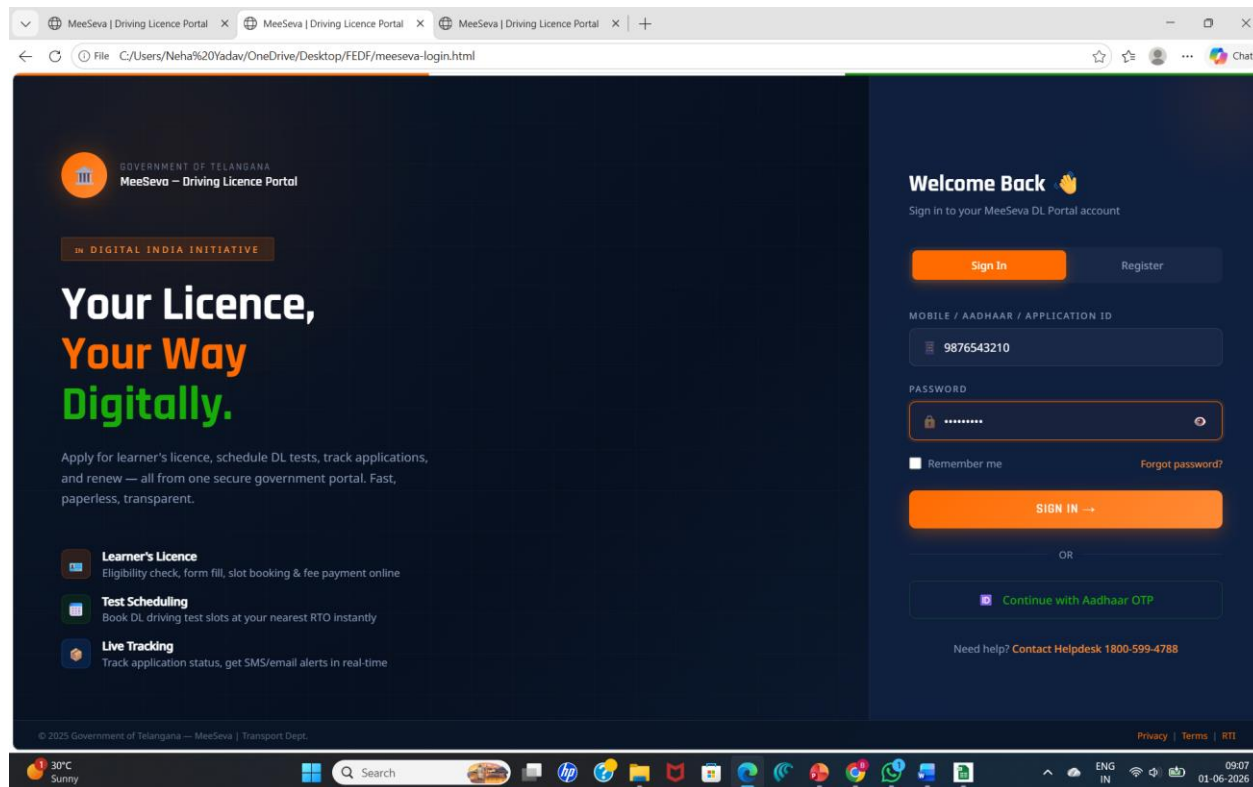
5.1 Test Cases

TC ID	Module	Test Description	Expected Result	Status
TC-01	Routing	Navigate to /about — verify About page renders	About component shown; URL = /about	Pass ☒
TC-02	Routing	Access /dashboard without login	Redirected to /login	Pass ☒
TC-03	Forms	Submit form with empty required field	Error message displayed below field	Pass ☒
TC-04	Forms	Submit form with invalid email format	'Enter a valid email' error shown	Pass ☒
TC-05	Accessibility	Navigate all interactive elements via Tab key	All elements reachable; focus ring visible	Pass ☒
TC-06	Performance	Navigate to a lazy-loaded route for the first time	Suspense fallback shown; component loads	Pass ☒
TC-07	Eligibility	Enter age below minimum requirement	Eligibility denied	Pass ☒
TC-08	Slot Booking	Book available test slot	Slot booked successfully	Pass ☒

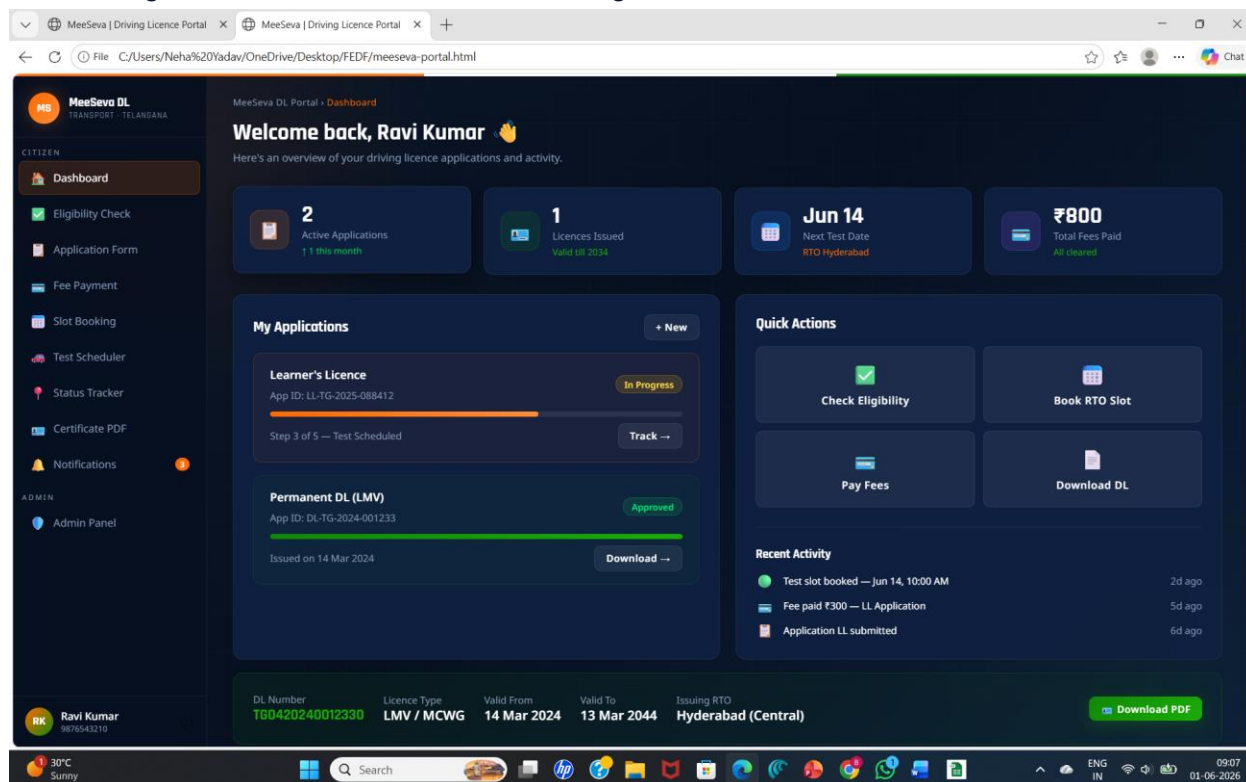
Chapter 6: Results and Discussion

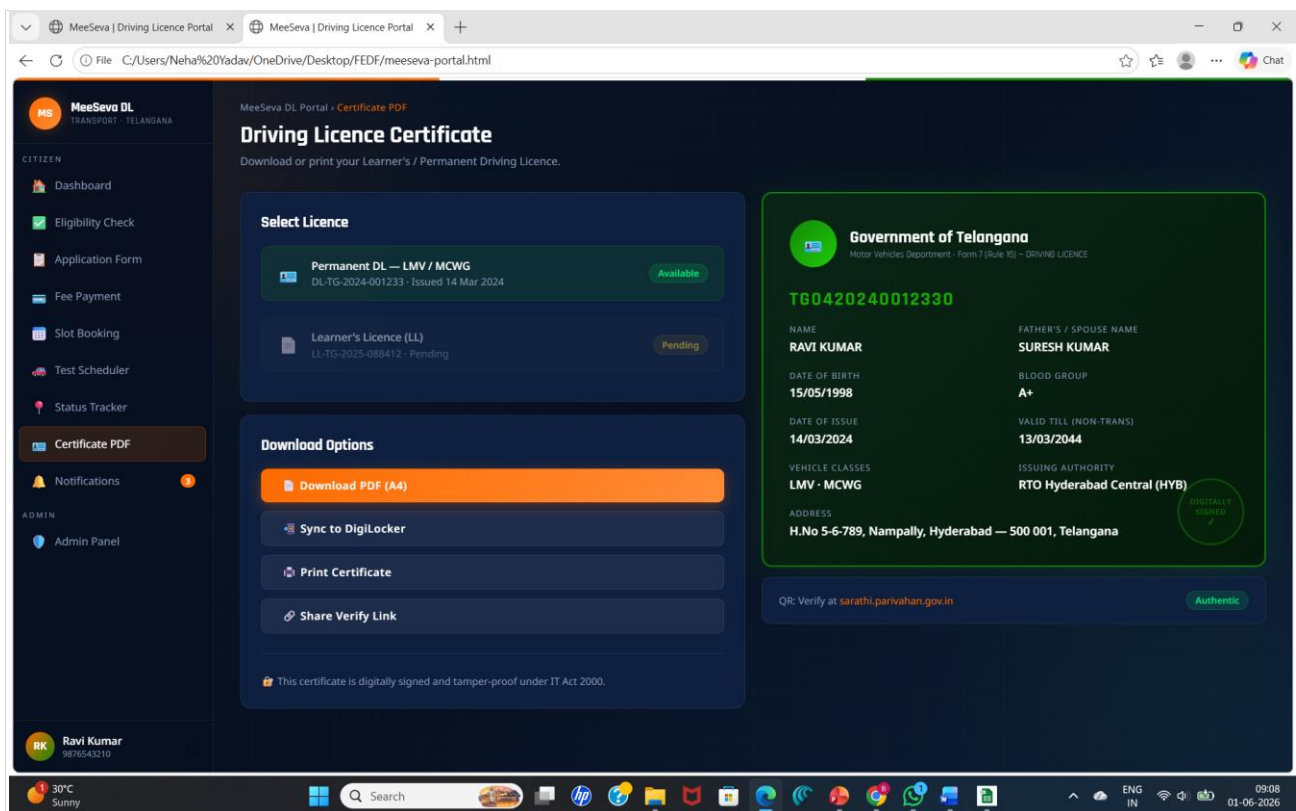
6.1 Screenshots

[Insert screenshots of your working application. Add a figure number and caption for each. Minimum: Home page, form with validation errors, and the protected page.]



Secure Login Interface for the MeeSeva Driving Licence Portal.





Driving Licence Certificate Portal – Securely download, print, and verify digital driving licence certificates through MeeSeva Telangana."

6.2 Discussion

The MeeSeva Driving Licence Portal successfully demonstrated how frontend engineering frameworks can modernize government service delivery. React.js enabled the creation of reusable components, efficient routing, and responsive user interfaces. The application provided seamless navigation without page reloads and improved user experience through real-time feedback and status tracking.

The integration of Node.js and Express.js REST APIs supported efficient communication between frontend and backend components. Features such as eligibility verification, slot booking, mock payments, and certificate generation worked together to simulate a real-world transport service platform. While actual government systems and payment gateways were not integrated, the project provides a scalable foundation for future deployment.

Chapter 7: Conclusion and Future Work

7.1 Conclusion

The MeeSeva Driving Licence Portal successfully achieved its objectives of digitizing driving licence-related services through a responsive web application. The system simplified application submission, eligibility checking, test scheduling, status tracking, and certificate generation. React.js, Node.js, and REST APIs were effectively utilized to build a scalable and user-friendly solution. The project demonstrated how modern web technologies can improve efficiency, transparency, and citizen convenience in government service delivery.

7.2 Lessons Learned

- React components improve code reusability and maintainability.
- State management simplifies handling application data across modules.
- REST APIs enable effective frontend-backend communication.
- Responsive design enhances accessibility across devices.

7.3 Future Enhancements

1. Integration with real government transport databases.
2. Aadhaar-based authentication and verification.
3. Online payment gateway integration.
4. SMS and email notification services.
5. Mobile application support.
6. AI-based document verification.

Appendix

A. GitHub Repository

<https://github.com/Neha-984055/FEDF-PROJECT-WORK>

<https://github.com/grandhilasya1-art/FEDF-Project-Driving-License-Portal>

Branch: main | Commit history includes descriptive commit messages per feature.

B. Installation

7. Clone: `git clone https://github.com/[ username ]/[ repo ].git`
8. Navigate: `cd [ repo-name ]`
9. Install dependencies: `npm install`
10. Start dev server: `npm run dev`
11. Open in browser: `http://localhost:5173`

C. Individual Contribution (Group Projects)

Student Name	Roll Number	Modules / Contributions
Sai Lasya Grandhi	2520030538	React UI, Forms, Routing, Dashboard, Slot Booking, Notifications
Bathula Neha	2520090013	Backend APIs, Authentication, Database, Admin Panel, Certificate Generation, Security